

SMART LOAD MONITORING SYSTEM

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Aim:

The Smart Load Monitoring System project involves designing a system that uses HX711 load cell sensor and Arduino Uno to monitor the load continuously and raise alarms whenever the load exceeds or is below the predetermined threshold value.

Statement of the problem:

Materials are stored in bulk quantities in warehouses, industries, and storage centers where their weight is constantly monitored. If the weight on the platform exceeds the permissible level, there will be overloading that can damage equipment and reduce safety.

Similarly, if the weight is less than a certain threshold, it could mean low levels of material or depletion which could interfere with the processes of production.

The traditional way of monitoring the weight of materials is through manual inspection carried out by the workers. This method is not only cumbersome but also error-prone since there is involvement of humans. In addition, since warehouses deal with many kinds of materials, manual monitoring cannot give accurate results regarding the current weight of the materials.

For the above problems, there is need for an automated Smart Load Monitoring System that measures the weight of the materials placed on a platform using an HX711 load cell sensor and processes the data via an Arduino Uno. The measured weight is then compared with the predetermined threshold values to identify whether the loading situation is normal, overloaded, or under-loaded.

If the measured weight surpasses the higher threshold value, an overload signal is triggered by the system, while in case the weight is below the lower threshold value, a low material signal is raised by the system.

This will enable the employees in the warehouse to take prompt action measures and increase efficiency. The suggested system offers constant monitoring, minimizes manual efforts, increases accuracy, and forms the base of future cloud-based monitoring systems.

Objectives:

1. Continuous monitoring of weight of the materials.
2. Measurement of weight using the HX711 load cell sensor.
3. Detection of low material level.
4. Detection of overloading condition.
5. Alerts generated from threshold values.
6. Simulation of the system using Wokwi.

Scope of the Solution:

The suggested solution consists of measuring the weight loaded on the platform through an HX711 load cell sensor continuously. Weight will then be processed by the Arduino Uno board and compared against pre-set threshold weights.

It offers the following features:

- Low Material Alert for weights less than 10 kg.
- Normal Load for weights between 10 kg and 40 kg.
- Overload Alert for weights more than 40 kg.

This solution can be further developed by integrating it with the cloud-based solutions like Blynk or ThingSpeak.

Required Components:

Hardware Components

1. Arduino Uno
2. HX711 Load Cell Sensor (50 kg)
3. Jumper Wires

Software Components

1. Wokwi Simulator
2. Arduino IDE
3. GitHub

Cloud Environment

1. Blynk Cloud (for future implementation)
2. ThingSpeak (optional)

Principle of Operation:

The load cell measures the force exerted on the weighing scale and generates an electrical signal output. The HX711 module amplifies the electrical signal generated by the sensor and sends the data to the Arduino Uno board.

The Arduino processes the data from the sensor, uses a calibration constant to measure the weight, and compares the weight to the defined thresholds.

If:

- $\text{Weight} < 10 \text{ kg} \rightarrow \text{LOW MATERIAL LEVEL}$
- $\text{Weight between } 10 \text{ kg and } 40 \text{ kg} \rightarrow \text{NORMAL LOAD LEVEL}$
- $\text{Weight} > 40 \text{ kg} \rightarrow \text{OVERLOAD LEVEL}$

Calibration:

The HX711 sensor generates uncalibrated digital data. This data needs to be calibrated to obtain kilogram readings.

Unbalanced readings:

0 kg = 0

10 kg = 4200

25.1 kg = 10542

50 kg = 21000

Calibration Factor:

$\text{kg} = \text{Unbalanced Reading} / 420$

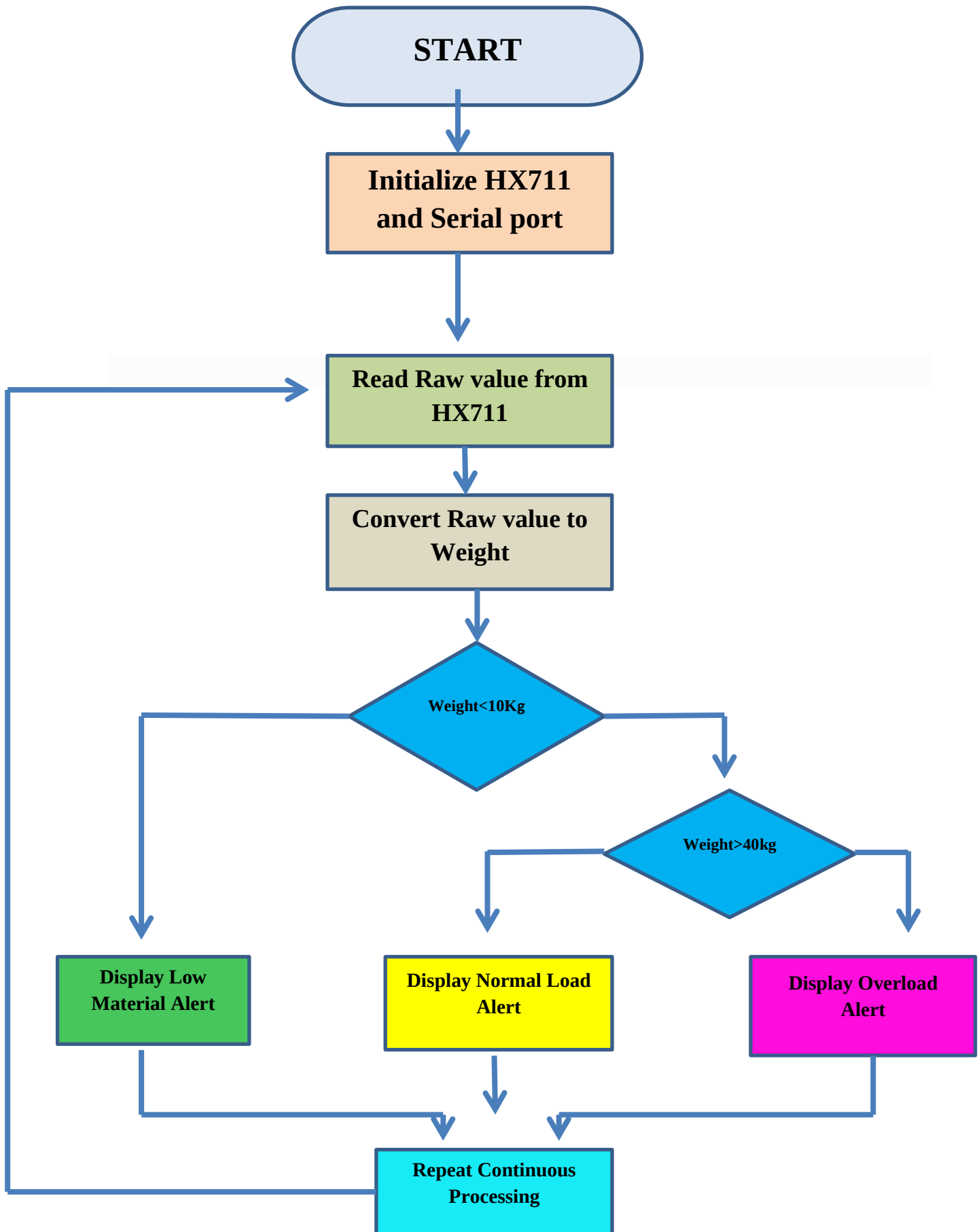
This calibration factor will be incorporated in the Arduino code.



Flowchart:

FLOWCHART OF SMART LOAD MONITORING SYSTEM

(USING HX711 LOAD CELL)



Arduino Programme Code:

```
#include "HX711.h"

#define DT 2
#define SCK 3

HX711 scale;

void setup() {
    Serial.begin(9600);
    scale.begin(DT, SCK);
}

void loop() {
    long rawValue = scale.read();
    float weight = rawValue / 420.0;

    Serial.print("Weight: ");
    Serial.print(weight);
    Serial.println(" kg");

    if(weight < 10) {
        Serial.println("LOW MATERIAL ALERT");
    }
    else if(weight > 40) {
        Serial.println("OVERLOAD ALERT");
    }
    else {
        Serial.println("NORMAL LOAD");
    }

    delay(1000);
}
```

Simulated Circuit:

This circuit was designed and tested with the help of Wokwi simulator.

Connections:

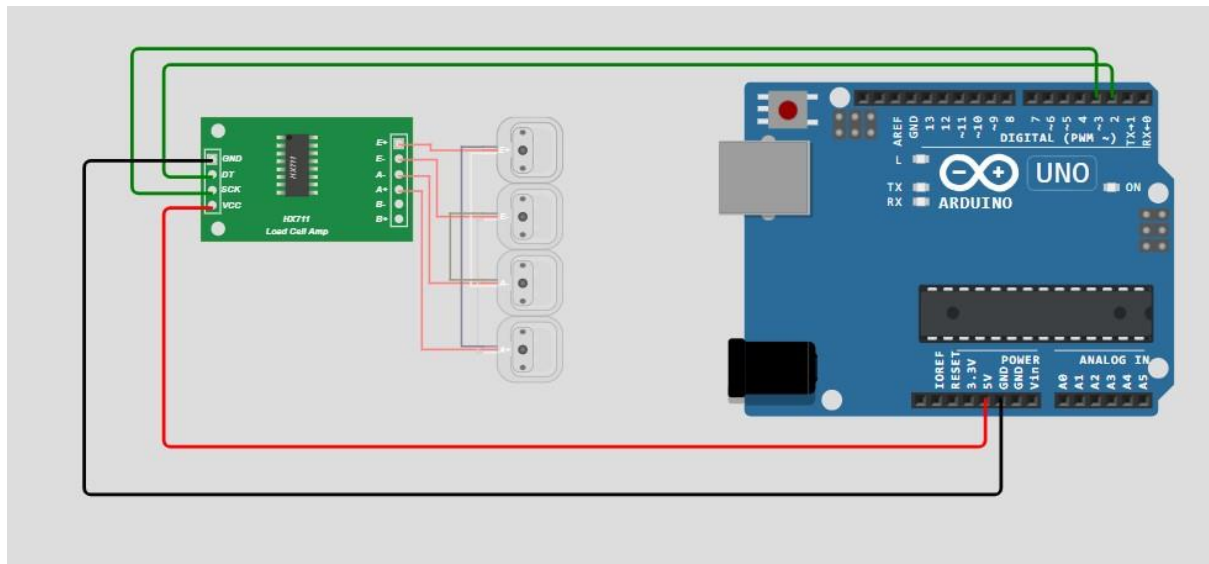
HX711 VCC to Arduino 5V

HX711 GND to Arduino GND

HX711 DT to Arduino D2

HX711 SCK to Arduino

Circuit Diagram



Results:

The Smart Load Monitoring System was able to measure the load and provide alarms according to the threshold set.

Results of the Test:

Weight = 5 kg → LOW MATERIAL ALARM

Weight = 25 kg → NORMAL LOAD

Weight = 45 kg → OVERLOAD ALARM

It has been successfully shown that weight can be monitored continuously through the HX711 load cell sensor.

Applications:

- Warehouses
- Industrial storage systems
- Inventory monitoring
- Smart logistics
- Material management systems
- IoT-based monitoring systems

Advantages:

- Continuous monitoring
- Precise weight measurements
- Ease of implementation
- Affordable
- Generation of real-time alerts
- Extensible to cloud platforms

Conclusion:

A successful design and simulation of the Smart Load Monitoring System using the HX711 load cell sensor on Wokwi is presented. The system provides continuous weight measurements and alerts in case of low material or overload situations. This project shows the fundamental approach to creating an IoT-based monitoring system and can be further developed with the integration of cloud services.